# Devashree Gupta

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## Summary

Fourth-year undergraduate studying Biomedical Engineering and Electrical Engineering, graduating May 2027. Interested in pursuing a career in medical devices, with a passion for neuroengineering and reproductive healthcare. Experience in sensor characterization, mixed-signal circuit design, embedded systems, and signal processing.

## Education

**Johns Hopkins University** **Baltimore, MD**

*B.S., Biomedical Engineering & B.S., Electrical Engineering (Dual Degree)* *Aug. 2023 – May 2027 (Expected)*

Cumulative GPA: 3.68

*Relevant Coursework:* Control Systems, Nonlinear Dynamics, FPGA Programming, Microfabrication Lab, Biological Models and Simulations, Digital Signal Processing, VLSI, Advanced Electronics Design

## Capstone Design Projects

**Airway Adaptor for Robotic Bronchoscopies** *Jan. 2025 – Present*

*Faculty Mentor: Dr. Elizabeth Logsdon | Clinical Mentor: Dr. Andrew Lerner, Sibley Memorial Hospital*

- Led prototyping for a robotic bronchoscopy attachment to keep visualization of the lungs unobstructed during multi-tool endotracheal intubation, allowing interventional pulmonologists to detect pulmonary hemorrhage early.
- Collaborated in an 8-person team, analyzed 500+ clinical studies, and shadowed pulmonologists in the operating room to derive user needs and design requirements.
- Modeled the device assembly in SolidWorks and fabricated 20+ design iterations by resin 3D printing, successfully testing the device in a mock OR.
- Presented at Johns Hopkins Design Day to 1,000+ attendees.

**Transperineal Biopsy Collection System** *Jan. 2024 – Aug. 2024*

*Faculty Mentor: Dr. Nick Durr | Industry Sponsor: Mendaera Inc.*

- Led verification and validation of a new transperineal biopsy system to improve prostate cancer diagnosis.
- Designed and built LVDT-instrumented fixtures to measure needle firing velocity, establishing the verification method for the firing mechanism's core design requirement.
- Completed a working biopsy-gun prototype designed against FDA and ISO requirements and validated it through a progression of low- to high-fidelity test methods.
- Presented at Johns Hopkins Design Day to 1,000+ attendees.

## Research Experience

**Department of Biomedical Engineering, Johns Hopkins University** *July 2025 – June 2026*

*Research Assistant, Engineering* *Baltimore, MD*

- Led electrical design of mosquito traps for attraction, capture, powder marking, and release for surveillance of *Aedes aegypti* mosquito populations in Brazil, Florida, and Costa Rica.
- Designed a relay control PCB in KiCAD and developed an Arduino-based embedded system for automated nightly operation.
- Isolated repeated field failures of the trap visual system to insufficient illumination and re-designed lighting to improve image quality.

**Comillas Pontifical University** *May 2025 – Aug. 2025*

*Signal Processing Engineering Intern* *Madrid, Spain*

- Built an RF signal-processing and classification pipeline to detect and identify drones by their RF signatures.
- Denoised raw RF captures with FFT-based filtering and wavelet transforms, improving SNR by 40%.
- Trained a convolutional neural network on the preprocessed signatures, reaching 88% classification accuracy.

**Brain Value Ltd.** *July 2024 – Sept. 2024*

*Data Science Intern* *Bangalore, India*

- Collected, processed, and analyzed biomarker and clinical trial datasets using machine learning methods; prepared reports and visualizations to characterize biomarker relationships.
- Processed clinical trial data, applying regression analysis, clustering, and predictive modeling using scikit-learn.

## Industry Experience

**Medtronic, Electrosurgical Generator Team** *June 2026 – Aug. 2026*
*Sensor Hardware Engineering Intern* *Lafayette, CO*

- Characterized the voltage and current sensing of a next-generation surgical generator across its full operating range to establish software calibration baselines, presented findings to 300+ Medtronic employees.
- Automated the full characterization loop in Python, sweeping loads, PWM drive modes, and harmonics on a self-built bench mount.
- Defined sensor gain and phase values from characterized data, delivering calibration baselines used in production firmware.
- Root-caused discrepancies between external probe and onboard measurements to unaccounted parasitic capacitance of traces.

**Johnson & Johnson MedTech, ECMO Team** *Jan. 2026 – June 2026*
*Product Development Engineering Intern* *Baltimore, MD*

- Developed test methods for cavitation and air handling in blood pumps, supporting next-generation pump design and IDE submission to the FDA, presented projects and findings to 100+ employees.
- Built end-to-end cavitation test protocol in accordance with ISO 18242, using a high-speed camera to detect cavitation formation at the impeller inlet, enabling repeatable compliance testing across builds.
- Investigated a de-airing method for blood pumps to prevent gas embolism, identifying air localization and modeling design changes for optimal air removal, documented findings in the DHF.
- Drafted and executed Test Method Validation protocols and tool specifications for critical components, applying Gage R&R and capability analysis using Minitab and a Keyence IM.

## Other Independent Projects

**CMOS Photosensor** *Feb. 2025 – May 2025*

- Designed a 2.5 mm × 2.5 mm CMOS image sensor in Cadence Virtuoso from schematic to verified layout.
- Integrated 3-transistor active pixel sensors with photodiodes, row shift registers, and column scan logic.
- Verified the completed layout with full DRC and LVS checks.

**Password-Encrypted Digital Keypad** *Feb. 2025 – May 2025*

- Built a passcode-locked keypad in discrete logic that unlocks only on a correct 7-digit entry sequence.
- Architected a Moore finite state machine with one-hot encoding across 8 sequential passcode-validation states.
- Prototyped in hardware with D flip-flops, logic gates, and a clock circuit, adding pull-down resistors to eliminate floating inputs and LED indicators for entry feedback.

**LED Music Sync Circuit** *Oct. 2024 – Dec. 2024*

- Designed an analog circuit that drives LEDs in sync with the frequency content of microphone audio, using three frequency bands.
- Unit-tested the electret-microphone amplification and filtering stages on a breadboard, then verified performance and tuned the frequency response in LTSpice simulation.

**BRAF V600E Mutation Study** *Feb. 2024 – Apr. 2024*

- Investigated how the BRAF V600E mutation alters MAPK/ERK signaling to drive uncontrolled proliferation.
- Extracted cellular metrics with CellProfiler, then compared oncogenic and wild-type populations in MATLAB using heatmaps, line graphs, and statistical analyses.
- Found significantly higher ERK activity, more cytoplasmic localization, and smaller nuclei in oncogenic cells.

## Teaching Experience

**Johns Hopkins University** *Jan. 2025 – Dec. 2025*
*Physics Learning Assistant* *Baltimore, MD*

- Reinforced electricity and magnetism concepts by leading group discussion and problem-solving sessions, in collaboration with professors and teaching assistants.

**Johns Hopkins University** *Aug. 2024 – Dec. 2025*
*First-Year Mentor* *Baltimore, MD*

- Supported incoming students via orientation activities, group events, and academic and campus resources.

**Johns Hopkins University**
*Design Cornerstone Teaching Assistant* *Baltimore, MD*

- Mentored first-year engineering teams through hands-on design and prototyping projects, advising on multidisciplinary design processes, electronics, Arduino hardware/software, and 3D printing.

- Helped troubleshoot student design challenges, provided feedback on prototypes and presentations, and collaborated with faculty to support course activities and maintain a safe prototyping environment.

## Technical Skills

- **Programming & Data Analysis:** Python (NumPy, SciPy, pandas, scikit-learn), C/C++, MATLAB, LabVIEW, NI DAQ, SQL, Git, Minitab
- **Circuits & Embedded Systems:** KiCAD, LTSpice, Cadence Virtuoso, Verilog, Arduino
- **CAD & Prototyping:** SolidWorks (incl. FEA Simulation), Rapid Prototyping, 3D Printing (SLA/FDM), Machining (CNC, Lathe, Weld, Mill, Laser Cutting)

## Leadership & Service

- **Professional Development Chair, AKDPhi (Asian-Interest Sorority):** Hosted resume workshops, networking events, and a headshot event.
- **Fundraising Chair, AKDPhi:** Raised $3,000 for breast cancer awareness through fundraising activities.
- **Hopkins Community Connection Advocate** (Jan. 2024 – May 2025): Assisted underserved patients and families at the Johns Hopkins Outpatient Center in overcoming barriers to social determinants of health.
- **Believe in Art Volunteer:** Volunteered at the Keswick Multi-Care Center, spending time and crafting with older patients living with dementia.

## Languages

English, Hindi, French, Marathi, Spanish